

Tommaso Mancuso

Full Stack Engineer & AI Solutions

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in LinkedIn Profile - Tommaso Mancuso

GitHub Profile - Tommaso Mancuso

Web Portfolio - Tommaso Mancuso

Professional Profile

Product-oriented Computer Engineer, specialized in full-stack software development and distributed architectures. Skilled in rapid prototyping through **AI-assisted coding** and in building AI-integrated solutions (RAG, LLM) for scalable SaaS systems and startups.

Work Experience

2025–2026 **Full Stack Engineer**, *Vendor Connect*, Remote (USA & Italy)

- **Core Architecture:** Designed and developed a scalable multi-tenant architecture for managing **Order-to-Cash (O2C)** flows.
- **Technical Refactoring:** Led the transition from a demo stage to a scalable production prototype.
- **Fintech Integrations:** Technical integration of external ecosystems and APIs to streamline the system's data acquisition processes.

2023–2024 **HMI & Software Developer**, *Creta Energie Speciali*, Calabria

- Developed **HMI** interfaces and control logic for the deep decarbonization of steel plants.
- Programmed real-time systems and virtual simulations in the steelmaking domain.

Featured Projects

2026 [🔗 Demo Mediterranea](#), *E-Commerce Prototype*, [🔗 Live Demo](#)

- **Browser-Only Architecture:** Multi-role e-commerce and back-office running entirely client-side — no database, API, or environment variables, with local persistence (**localStorage/sessionStorage**).
- **Full O2C Flow:** Replicates the real order lifecycle (submitted → approved → paid → shipped → received).

2026 [🔗 Company Document Portal](#), *Web Applications II*, Politecnico di Torino

- **Distributed Backend:** Microservices ecosystem in **Spring Boot / Kotlin** with the **TUS** protocol for resilient, chunked large-file uploads.
- **Event-Driven Architecture:** Implemented the **Transactional Outbox Pattern** with Debezium, an **Apache Kafka** broker, and hybrid persistence (**PostgreSQL + pgvector** and **MinIO** as S3 Object Storage).
- **Zero-Trust Security:** Centralized identity management via **Keycloak (IAM)** with OAuth2/OIDC flows and a dedicated API Gateway.
- **Retrieval-Augmented Generation Pipeline (Spring AI)** integrated with retrieval filters based on user permissions (ACL). Advanced testing via **Testcontainers** (Docker).

2026 [🔗 Last Ride \(SPA Web Game\)](#), *Web Applications I*, Politecnico di Torino

- **Built a Single Page Application** in **React / Vite** with a finite-state machine for real-time management of the game's setup, planning, and execution flows.
- **Real-Time Server-Side Validation** in **Node.js / Express**. Handled routing over metropolitan network graphs, path validation, and event processing stored in **SQLite**.
- Managed asynchronous timers with auto-submit mechanisms and sequential rendering of the trip stages.

2026 [🔗 Foresteria - Il Cacciatore](#), *AI-Driven SaaS Prototype*, [🔗 Live Demo](#)

- **Multi-role management system** with a **revenue** dashboard and real-time occupancy monitoring, built in **Next.js**.
- **Integrated Logistics:** Guest registry and housekeeping workflow modules, developed with **AI-first** logic (Supabase backend).

2025 [🔗 Voyago - Mobile Application](#), *Android SDK*

- Trip Manager app built in Kotlin with an **MVVM** architecture and **Firebase Firestore**, including social interaction features.

2025 [🔗 Rip Hit](#), *AI Game*, *Unity*, *C#*

- Narrative beat 'em up powered by an **LLM (Gemini)**. Built the Combat System, Event Engine, and the AI integration driving an in-game Character.

Education

Ongoing **MSc in Digital Media Engineering**, *Politecnico di Torino*

Focus: Distributed Systems, Machine Learning (ResNet-50), Advanced Web Dev, Interactive Systems.

2023 **BSc in Computer Engineering**, *Politecnico di Torino*

Final thesis focused on Asymmetric Cryptography. Focus: Linux Operating Systems, SQL, Data Structures.

Technical Skills

Backend Core Kotlin, Spring Boot, Apache Kafka, Node.js, Express, Keycloak (IAM), TUS Protocol

AI & Frontend Spring AI, Google Vertex AI, RAG (pgvector), Next.js, React, Supabase, TypeScript

DevOps & Infra Docker/Compose, Testcontainers, MinIO (S3), OpenTelemetry, Loki, Prometheus, Grafana

Core Skills SQL, SQLite, PostgreSQL, Git, GitHub, Linux Shell, Distributed Systems Design

Languages English - **C1**, Italian - **Native**